



**CITY ENGINEER'S OFFICE**  
2101 O'Neil Avenue  
Cheyenne, Wyoming 82001

A COMMUNITY OF CHOICE

## MEMO

**TO:** Honorable Mayor Marian J. Orr, City of Cheyenne  
Members of the City Council, City of Cheyenne

**FROM:** Wesley D. Bay, P.E., Deputy City Engineer

**DATE:** January 10, 2019

**SUBJECT:** Agreement for Aerial Photography and LIDAR Service, Laramie County,  
Wyoming.  
2019 Aerial Photography and LiDAR Project.

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Please find the attached contract for the above referenced Agreement. This Agreement outlines the terms and conditions of an agreement between the City of Cheyenne (on behalf of the Cheyenne Laramie County GIS Cooperative) and Aero-Graphic, Inc. on the gathering of aerial imagery and LiDAR data for use with the City's and Laramie County's GIS system. The total cost of the project is \$130,000. This will be paid for jointly by the City, the Board of Public Utilities (BOPU), Laramie County, and the Cheyenne Metropolitan Planning Organization (MPO). The City will be contributing \$38,452.

Approved as to  
form only:

LS

Date: 1-9-19

**AGREEMENT**  
**for**  
**AERIAL PHOTOGRAPHY AND LIDAR SERVICE**  
**LARAMIE COUNTY, WYOMING**  
**2019 Aerial Photography and LiDAR Project**  
**2,683 Sq. Miles**

THIS AGREEMENT is between the City of Cheyenne (hereinafter referred to as Owner) on behalf of the Cheyenne Laramie County GIS Cooperative and Aero-Graphics, Inc. (hereinafter referred to as Provider);

WITNESSETH:

WHEREAS, Owner intends to have Provider perform aerial photography and LiDAR services for the 2019 Aerial Imagery and LiDAR Project (hereinafter referred to as the Project); and,

WHEREAS, Owner requires certain professional services in connection with the Project (hereinafter referred to as the Services); and,

WHEREAS, Provider is prepared to provide such Services;

NOW THEREFORE, in consideration of the promises contained herein, the parties hereto agree as follows:

**ARTICLE 1 - ARTICLE EFFECTIVE DATE**

The effective date of this Agreement shall be February 1, 2019 or when fully executed.

**ARTICLE 2 - SCOPE OF WORK AND TIME FOR PERFORMANCE**

Provider shall perform the Services described in the Scope of Work attached hereto as Attachment A, which is incorporated by reference as part of this Agreement. Provider shall deliver all data collected pursuant to Attachment A to the Owner by December 31, 2019.

**ARTICLE 3 - COMPENSATION**

In consideration of the Services to be performed pursuant to this Agreement, the Provider will bill the Owner and the Owner agrees to pay Provider a not to exceed amount of One Hundred Thirty Thousand Dollars (\$130,000.00). Payment shall be based on work completed, delivered and accepted by the Owner in accordance with the price quote contained in Attachment A, and incorporated by reference as part of this agreement. Payment shall not be made for any rejected photography or LiDAR data. Payment shall constitute full compensation for all work, defrayment of all fees and other costs. Provider shall submit invoices to Owner upon completion of each scheduled phase of the project as follows: 1) ground control survey services (approximately \$10,000); 2) aerial photography acquisition (approx. \$36,400); 3) LiDAR acquisition (approx. \$54,600); 4) imagery orthorectification and delivery (approx. \$11,600); and 5) LiDAR processing

and delivery (approx. \$17,400). Payment to Provider shall be made within 60 days of each invoice date. A 1.8% per month interest penalty will apply to any unpaid balance dating back to the original invoice date after 60 days.

#### **ARTICLE 4 - STANDARD OF CARE**

Provider shall exercise the same degree of care, skill, and diligence in the performance of the Services as is ordinarily provided by professional aerial photography and LiDAR firms under similar circumstances and Provider shall, at no cost to Owner, re-perform services which fail to satisfy the foregoing standard of care.

#### **ARTICLE 5 - LIABILITY AND INDEMNIFICATION**

5.1 General. Having considered the risks and potential liabilities that may exist during the performance of the Services and in consideration of the promises included herein, Owner and Provider agree to allocate such liabilities in accordance with this Article 5. Words and phrases used in this Article shall be interpreted in accordance with customary insurance industry usage and practice.

5.2 Indemnification. Provider shall indemnify and defend Owner from any and all legal liability for damages arising out of the performance of the Services for Owner.

5.3 Defense of Claims. In the event an action for damages is filed in which negligence is alleged on the part of both Owner and Provider, Provider agrees to defend Owner. In the event any joint and several judgment is rendered against Owner and Provider, Provider agrees to indemnify Owner for Owner's share of damages attributed to Owner.

5.4 Employee Claims. Provider shall indemnify Owner against legal liability for damages arising out of claims by Provider's employees.

5.5 Other Project Indemnities. Indemnity provisions, similar to those set forth herein, shall be incorporated into all project contractual arrangements entered into by Owner and shall protect Owner and Provider to the same extent.

5.6 Survival. Upon completion of all Services, obligations and duties provided for in this Agreement, or in the event of termination of this Agreement for any reason, the terms and conditions of this Article shall survive.

#### **ARTICLE 6 - INDEPENDENT CONTRACTOR**

Provider undertakes performance of the Services as an independent contractor and shall be wholly responsible for the methods of performance. Owner shall have no right to supervise the methods used but Owner shall have the right to observe such performance and require performance of Provider obligations hereunder. Provider shall cooperate with Owner in performing Services under this agreement.

## **ARTICLE 7 - COMPLIANCE WITH LAWS**

In performance of the Services, Provider will comply with applicable regulatory requirements including federal, state, and local laws, rules, regulations, orders, codes, criteria and standards. Provider shall procure the permits, certificates, and licenses necessary to allow Provider to perform the Services.

## **ARTICLE 8 – INSURANCE**

During the performance of the Services under this Agreement, Provider shall maintain the following insurance coverage from insurance companies having a minimum A.M. Best rating of A- (excellent):

1. General liability insurance. For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.
2. Automobile liability insurance with limit no less than \$1,000,000 per accident for bodily injury and property damage.
3. Worker's Compensation Insurance in accordance with statutory requirements and Employer's Liability Insurance with limits of not less than \$1,000,000 for each occurrence.

Provider shall name Owner as an Additional Insured by endorsement on its general liability insurance policies prior to beginning any work under this Agreement and shall provide Owner with a copy of the policy declaration and endorsement pages. Provider shall likewise furnish Owner certificates of insurance acknowledging the above-stated coverages. It is understood and agreed that these policies are primary and not contributory.

## **ARTICLE 9 - OWNER'S RESPONSIBILITIES**

Owner shall be responsible for furnishing Provider with existing and previous ground control coordinates and LiDAR and ortho surfaces to expedite data processing.

## **ARTICLE 10 - REUSE OF DOCUMENTS**

All documents, including drawings, specifications and computer software, prepared by Provider pursuant to this Agreement are instruments of service in respect to this Project. They are not intended or represented to be suitable for reuse by Owner or others on extensions of this Project or on any other project. Any reuse without written verification or adaptation by Provider for the specific purpose intended will be at Owner's sole risk and without liability or legal exposure to Provider.

## **ARTICLE 11 - TERMINATION OF CONTRACT**

Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

In the event the Provider fails to strictly perform in accordance with this Agreement, the Owner may elect to correct the deficiencies and charge the Provider. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party, each party shall bear its own costs and expenses, including without limitation, attorney's fees.

Owner shall have the right to terminate this Agreement or suspend performance thereof for Owner's convenience upon written notice to Provider and Provider shall terminate or suspend performance of Services on a schedule acceptable to Owner and shall deliver to the Owner within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Provider in performing this Agreement, whether completed or in process. In the event of termination or suspension for Owner's convenience, Owner shall pay Provider for all accepted Services performed. Such acceptance shall not be unreasonably withheld.

## **ARTICLE 12 - NONDISCLOSURE OF PROPRIETARY INFORMATION**

Provider shall consider all information provided by Owner to be proprietary unless such information is available from public sources. Provider shall not publish or disclose proprietary information for any purpose other than the performance of the Services without the prior written authorization of Owner or in response to legal process.

## **ARTICLE 13 - NOTICE**

Any notice, demand, or request required by or made pursuant to this Agreement shall be deemed properly made if personally delivered in writing or deposited in the United States mail, postage prepaid, to the address specified below.

To Provider: Aero-Graphics  
Bernie Doud, Project Manager  
40 W Oakland Ave  
Salt Lake City UT 84115

To Owner: Cheyenne Laramie County GIS Cooperative  
Jordan Evans, GIS Coordinator  
310 W 19<sup>th</sup> St, Room 410  
Cheyenne, WY 82001

Nothing contained in this Article shall be construed to restrict the transmission of routine communications between representatives of Provider and Owner.

#### **ARTICLE 14 - UNCONTROLLABLE FORCES**

Neither Owner nor Provider shall be considered to be in default of this Agreement if delays in or failure of performance shall be due to uncontrollable forces the effect of which, by the exercise of reasonable diligence, the non-performing party could not avoid. The term "uncontrollable forces" shall mean any event which results in the prevention or delay of performance by a party of its obligations under this Agreement and which is beyond the control of the non-performing party. It includes, but is not limited to, fire, flood, earthquakes, storms, lighting, epidemic, war, riot, civil disturbance, sabotage, inability to procure permits, licenses, or authorizations from any state, local, or federal agency or person for any of the supplies, materials, accesses, or services required to be provided by either Owner or Provider under this Agreement, strikes, work slowdowns or other labor disturbances, and judicial restraint.

Neither party shall, however, be excused from performance if nonperformance is due to uncontrollable forces which are removable or remediable and which the nonperforming party could have, with the exercise of reasonable diligence removed or remedied with reasonable dispatch. The provisions of this Article shall not be interpreted or construed to require Provider or Owner to prevent, settle, or otherwise avoid a strike, work slowdown, or other labor action. The non-performing party shall, within a reasonable time of being prevented or delayed from performance by an uncontrollable force, give written notice to the other party describing the circumstances and uncontrollable forces preventing continued performance of the obligations of this Agreement.

#### **ARTICLE 15 - GOVERNING LAW**

The parties mutually understand and agree this Agreement shall be governed by and interpreted pursuant to the laws of the State of Wyoming. If any dispute arises between the parties from or concerning this Agreement or the subject matter hereof, the claims procedure set forth in the Wyoming Governmental Claims Act, Wyo. Stat. § 1-39-113, shall apply, and any suit or proceeding at law or in equity shall be brought in the District Court of the State of Wyoming, First Judicial District, sitting at Cheyenne, Wyoming.

#### **ARTICLE 16 – MISCELLANEOUS**

16.1 Nonwaiver. A waiver by either Owner or Provider of any breach of this Agreement shall not be binding upon the waiving party unless such waiver is in writing. In the event of a

written waiver, such a waiver shall not affect the waiving party's rights with respect to any other or further breach.

**16.2 Severability.** The invalidity, illegality, or unenforceability of any provision of this Agreement, or the occurrence of any event rendering any portion or provision of this Agreement void shall in no way affect the validity or enforceability of any other portion or provision of the Agreement unless provision is expressly stated to be of the essence of this Agreement. If such provision is so identified, then a determination of its invalidity shall cause the entire Agreement to be void.

## **ARTICLE 17 - INTEGRATION AND MODIFICATION**

This Agreement represents the entire and integrated agreement between the Parties and supersedes all prior negotiations, representations, or agreements, either written or oral. This Agreement may be amended only by a written instrument signed by each of the Parties.

## **ARTICLE 18 - SUCCESSORS AND ASSIGNS**

Owner and Provider each binds itself and its directors, officers, partners, successors, executors, administrators, assigns and legal representatives to the other party to this Agreement and to the partners, successors, executors, administrators, assigns, and legal representatives of such other party, in respect to all covenants, agreements, and obligations of this Agreement.

## **ARTICLE 19 - ASSIGNMENT**

Neither Owner nor Provider shall assign, sublet, or transfer any rights under or interest in (including, but without limitation, moneys that may become due or moneys that are due) this Agreement without the written consent of the other, except to the extent that the effect of this limitation may be restricted by law. Unless specifically stated to the contrary in any written consent to an assignment, no assignment will release or discharge the assignor from any duty or responsibility under this Agreement.

Nothing contained in this paragraph shall prevent Provider from employing such independent consultants, associates, and subcontractors as he may deem appropriate to assign him in the performance of the Services hereunder.

## **ARTICLE 20 - THIRD PARTY RIGHTS**

Nothing herein shall be construed to give any rights or benefits to anyone other than Owner and Provider.

## **ARTICLE 21 – SOVEREIGN IMMUNITY**

Owner does not waive their Governmental/Sovereign Immunity, as provided by any applicable law including W.S. § 1-39-101 *et seq.*, by entering into this Agreement. Further, Owner

fully retains all immunities and defenses provided by law with regard to any action, whether in tort, contract or any other theory of law, based on this Agreement.

IN WITNESS WHEREOF, the parties have hereunto set their hands and seals the day and year indicated.

CITY OF CHEYENNE, WYOMING

By: \_\_\_\_\_  
Marian J. Orr, MAYOR

Date: \_\_\_\_\_

(SEAL)

ATTEST:

\_\_\_\_\_  
Kristina F. Jones, City Clerk

Aero-Graphics, Inc.

\_\_\_\_\_

By: Bernie Doud\_\_\_\_\_

Title: Project Manager\_\_\_\_\_

Date: 8 January 2019\_\_\_\_\_



## **Attachment A – Scope of Work**

### **Ground Control**

- Ground control survey, including LiDAR-specific targets, to support the aerial photography and LiDAR deliverables listed below. Approximately 22 points, will recover and refresh, or replace as necessary, points from the 2014 flight. The image files shall be geoprocessed and delivered in Wyoming State Plane East Zone, NAD83 (CORS96) U.S. survey feet.

### **Aerial Photography**

- Vexcel UltraCam Eagle Digital Image acquisition at 6" GSD
- GNSS/IMU trajectory post-processing
- Radiometric processing of stereo imagery; samples will be submitted for approval of color balance, density, contrast, etc.
- Analytical Aerotriangulation
- Tilt will not exceed four degrees for any photographic frame and will average not more than two degrees for any ten consecutive frames. Relative tilt exceeding six degrees between any two successive frames may be cause for rejecting that portion of the flight lines.
- Crab angle as measured from the average line of flight will not exceed five degrees. The course heading differential between any two successive exposures will not exceed five degrees.
- The project area will be collected when the sun angle is not less than 40°. On a cloud free or overcast day. No cloud shadows should be apparent.
- RGBI orthorectified imagery at 0.5' GSD in LZW Compressed TIFF/TFW and overall SID/SDW formats.
- Cells in tiles with no collected data shall be given a NoData value.
- Acquisition specifications and flight plan included below

Aerial Photography acquisition is to be undertaken in snow-free, leaf-off conditions. Owner and Provider will consult together and agree on an acceptable acquisition timeframe given the conditions that present themselves in Spring 2019. An alternate flying season in Fall 2019 may be considered if conditions are not conducive to a Spring acquisition.

### **LiDAR**

- Optech Galaxy PRIME LiDAR acquisition at 2ppsm
- GNSS/IMU trajectory post-processing
- LiDAR data calibration to LAS 1.2
- Bare earth data classification to LAS 1.2
- Control (absolute) accuracy report
- Technical Project Report detailing acquisition, processing, methodology, and results
- Acquisition specifications and flight plan included below

LiDAR acquisition may be undertaken any time after execution of this contract when snow-free conditions are present and when clouds are not between the aircraft and the ground. Lingering snow on north faces and in tree shadows may be acceptable but approval of Owner is required.

**Final Delivery of all data will occur by December 31, 2019.**

**Total Price      \$130,000.00**

**Bid #4437 Cheyenne/Laramie - UCE v3**

v3

|                    |       |
|--------------------|-------|
| Altitude (ft, AGL) | 9,600 |
| GSD                | 6"    |
| OL(%) average      | 60    |
| SL(%) average      | 30    |
| Lines              | 52    |
| Images             | 4,252 |

**Bid #4437 Cheyenne/Laramie - Galaxy/P1**

|                                    |       |
|------------------------------------|-------|
| Altitude (ft, AGL)                 | 5,200 |
| Speed(kts) max                     | 120   |
| PRF(kHz)                           | 200   |
| PIA (Pulses in air)                | 3     |
| Laser Power                        | High  |
| Lines                              | 98    |
| Scan Freq(Hz)                      | 46    |
| Scan Angle (Full)                  | 46    |
| Planned Swath Trim (Total Degrees) | 1     |
| PPM^2                              | 2.21  |
| Cross Track Res (m)                | 0.67  |
| Down Track Res (m)                 | 0.67  |
| LiDAR Sidelap %                    | 20    |
| LIDAR Swath Width (m)              | 1,358 |



UltraCam Eagle 6" GSD Flight Plan  
52 Flightlines; 4,252 Exposures



Optech Galaxy PRIME 2PPSM Flight  
Plan  
98 Flightlines

